WORKSHEET

# Chapter 7: Inverse trigonometric functions

**1** Find the following.

**a** $\frac{d}{dx}\sin^{-1}(x^2)dx$

**b** $\frac{d}{dx}\tan^{-1}(3x-4)dx$

**c** $\frac{d}{dx}\cos^{-1}(3x+7)$

**d** $\frac{d}{dx}\tan^{-1}[4\ln(x)]$

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**2** Find the following.

**a** $\int \frac{1}{1+9x^2}dx$

**b** $\int \frac{1}{\sqrt{1-49x^2}}dx$

**c** $\int \frac{e^x}{1+e^{2x}}dx$

**d** $\int \frac{-3}{\sqrt{1-9x^2}}dx$

**3** Find the following.

**a** $\int \frac{dx}{\sqrt{25-4x^2}}$

**b** $\int \frac{dx}{\sqrt{9-16x^2}}$

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**c** $\int \frac{-1}{\sqrt{16-9x^2}}dx$

**d** $\int \frac{dx}{36+4x^2}$

**e** $\int \frac{dx}{2+9x^2}$

**f** $\int \frac{dx}{\sqrt{3-x^2}}$

**4** Calculate the exact values of the following.

**a** $\int_{-1}^{1} \frac{1}{1+x^2}dx$

**b** $\int_{\frac{-1}{2}}^{\frac{1}{2}} \frac{1}{\sqrt{1-x^2}}dx$

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**c** $\int_0^{\frac{\sqrt{3}}{2}} \frac{1}{\sqrt{3-x^2}}dx$

**d** $\int_0^{4\sqrt{3}} \frac{1}{16+x^2}dx$

**5** Calculate the following (correct to three significant figures where necessary).

**a** $\int_{-1}^{1} \frac{dx}{\sqrt{36-9x^2}}$

**b** $\int_0^1 \frac{-1}{\sqrt{12+4x^2}}dx$

**c** $\int_{\frac{1}{2}}^{1} \frac{dx}{24+9x^2}$

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**d** $\int_{-1}^{1} \frac{dx}{\sqrt{25 - x^2}}$

**6** Consider the integral $\int \frac{dx}{\sqrt{5 + 4x - x^2}}$.

**a** Complete the square of the expression in the radical sign so that it takes the form $a^2 - u^2$.

**b** Use the result obtained in part **a** to show that $\int \frac{dx}{\sqrt{5 + 4x - x^2}} = \arcsin\left(\frac{x - 2}{3}\right) + c$.

**7** Use the same process as outlined in Question **6** to show that:
$\int \frac{dx}{x^2 - 4x + 7} = \frac{1}{\sqrt{3}} \arctan\left(\frac{x - 2}{\sqrt{3}}\right) + c$.

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## Answers

**1** **a** $\dfrac{2x}{\sqrt{1-x^2}}$

**b** $\dfrac{3}{1+(3x-4)^2}$

**c** $\dfrac{-3}{\sqrt{1-(3x+7)^2}}$

**d** $\dfrac{4}{x\left\{1+16\left[\ln(x)\right]^2\right\}}$

**2** **a** $\dfrac{1}{3}\arctan(3x)+c$

**b** $\dfrac{1}{7}\arcsin(7x)+c$

**c** $\arctan(e^x)+c$

**d** $\arccos(3x)+c$

**3** **a** $\dfrac{1}{2}\arcsin\left(\dfrac{2x}{5}\right)+c$

**b** $\dfrac{1}{4}\arcsin\left(\dfrac{4x}{3}\right)+c$

**c** $\dfrac{1}{3}\arccos\left(\dfrac{3x}{4}\right)+c$

**d** $\dfrac{1}{12}\arctan\left(\dfrac{x}{3}\right)+c$

**e** $\dfrac{1}{3\sqrt{2}}\arctan\left(\dfrac{3x}{\sqrt{2}}\right)+c$

**f** $\arcsin\left(\dfrac{x}{\sqrt{3}}\right)+c$

**4** **a** $\dfrac{\pi}{2}$

**b** $\dfrac{\pi}{3}$

**c** $\dfrac{\pi}{6}$

**d** $\dfrac{\pi}{12}$

**5** **a** $\dfrac{\pi}{9}$

**b** $-0.275$

**c** $0.0172$

**d** $0.403$

**6** **a** $5+4x-x^2=3^2-(x-2)^2$

**b** Demonstration.

**7** Use $x^2-4x+7=(x-2)^2+\left(\sqrt{3}\right)^2$.

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